Homework solutions 3/31/26

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1 Nikhil's notes, section 9.1

1. Assume the contrary. Then there exists $v \in H$ with degree in G greater than degree in H. Therefore there must exist $w \in G$ such that $\{v, w\}$ is an edge in G but not in H. But then H is not maximal – contradiction.

2. First, assume that $H = H_i$ (in other words, there are no other connected components in H). Then every vertex of H, and therefore every vertex of G, and therefore every vertex of C, must lie in H_i. Now, assume that H has other connected components. Then, since G was connected, there must exist some edge in G between some vertex in H_i and some vertex in some other connected component of H. This edge must have been removed when computing $H = G - E(C)$, so it must be an edge in C. Since H_i contains an endpoint of an edge in C, it contains a vertex of C.

3. The only connected graphs with six vertices and five edges are trees (HW 8.1.4). Every tree has a leaf, which has degree 1, so it cannot have an Euler circuit.

4. Such a graph does exist. This solution will explain one way of finding it. If such a graph exists, then all six of its vertices must have even degree – let us say that they are $2d_1, \ldots, 2d_6$. Then $2d_1 + \cdots + 2d_6 = 2 \times 7$, since there are seven edges, or $d_1 + \cdots + d_6 = 7$. Since each d_i is greater than 0, most of them must be equal to 1, while one must be equal to 2. In other words, the graph must have six vertices of degree 2 and one vertex of degree 4. This suggests a graph made of two cycles – a cycle with three vertices and a cycle with four turn out to be the right numbers – that share a single vertex, for a total of $3 + 4 = 7$ edges and $3 + 4 - 1 = 6$ vertices. The Euler circuit is given by making a circuit around one cycle, starting and finishing at the shared vertex, and then making another around the other cycle.

2 Rosen, section 6.1

1. There are 10^3 ways to choose three digits and 26^3 ways to choose three letters, so there are $10^3 \times 26^3 + 26^3 \times 10^3 = 2 \times 10^3 \times 26^3$ ways to make the license plates.

2. Following 6.1.2, example 7, the answer to part (b) is (5)(4)(3)(2)(1) and the answer to part (c) is (6)(5)(4)(3)(2).

3. Following 6.1.2, example 6, there are 2^n such functions.

4. There are 2^{100} subsets, of which one is empty and 100 have exactly one element, so the answer is $2^{100} - 101$.

5. If n is even, this is equivalent to asking how many bitstrings there are of length $n/2$ (since we can pick any such bitstring, reverse it, and combine the two to get a palindrome of length 2). There are $2^{n/2}$. If n is odd, we can get any palindrome of length n by taking a palindrome of length $n-1$ and then sticking either a 1 or a 0 at the center position, since that position remains the same when we reverse the string. There are therefore $2 \times 2^{(n-1)/2} = 2^{(n+1)/2}$ palindromes.

6. Let us first consider the number of ways to seat four people in a line. This is (10)(9)(8)(7). Now, if we wrap that line into a circle, and consider two seatings the same when everyone has the same neighbors, that is equivalent to saying that the seating is unchanged if we move everyone around the circle to the left or the right by the same number of places. There are four ways to do this (leave people alone, move people one to the right, move people two to the right, move people three to the right). Therefore the total number of possible seatings is $\frac{(10)(9)(8)(7)}{4}$.

7. First, consider the strings with five (or more) consecutive zeros. Let k be the position of the first zero in this group of five or more. If $k = 1$, the string looks like $00000xxxxx$. There are therefore 2^5 possibilities, since there are five blank spaces that can be filled by zeros or ones. If $k = 2$, the string looks like $100000xxxx$, for 2^4 possibilities. If $k = 3$, the string looks like $x100000xxx$, again for 2^4 possibilities. If $k = 4$, the string looks like $xx100000xx$, again for 2^4 possibilities, and, if $k = 5$, the string looks like $xxxx100000$, again for 2^4 possibilities. So there are $2^5 + 5 \times 2^4 = 112$ possible strings of length 10 with five or more consecutive zeros. Similarly, there are 112 strings with five or more consecutive ones. But the two strings 0000011111 and 1111100000 have both, so the total number of strings with five or more zeros or five or more ones is $112 + 112 - 2 = 222$.

3 Rosen, section 6.3

1. Following 6.3.3, example 15, the solutions are, in order, $C(12,3)$, $C(12,0)+C(12,1)+C(12,2)+C(12,3)$, $2^{12} - (C(12,0) + C(12,1) + C(12,2) +$

$C(12,3)$), and $C(12,6)$ (since, if a bit string of length 12 contains an equal number of 0s and 1s, it must contain six of each.

2. Following 6.3.2, example 7, we note that part (a) is equivalent to finding permutations of 7 items (of which there are 7!) and part (c) is equivalent to finding the 5! permutations of 5 items. Part (e) requires finding permutations which contain the strings CAB and BED, which is the same as finding permutations that contain the single string $CABED$. There are three letters in $ABCDEFGH$ not contained in this five-letter string, so the number is the number of permutations of four elements: 4!.

3. We first position the women. There are 10! ways of doing this. We also order the men from left to right. There are 6! ways of doing this. Now, there are eleven spaces where a man could stand: one between each pair of women, one on the left of the line, and one on the right. We have to choose six of them to ensure that no men are standing next to each other. The total number of possibilities is therefore $10! \times 6! \times C(11,6)$.

4. The trick here is to consider how many ways to form the committee there are if it has exactly k men. This is the number of ways to choose k men from ten and $6-k$ women from fifteen, or $C(10,k) \times C(15,6-k)$. Call this function $f(k)$. The requirement that there are more women than men is satisfied exactly when $k = 0$, 1, or 2, so the total number of ways to form the committee is $f(0) + f(1) + f(2)$.